Specialization of a Large Language Model for Question Answering on the Rules of Magic: The Gathering

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Abstract

Large Language Models (LLMs) have reshaped the landscape of artificial intelligence by demonstrating strong general-purpose linguistic capabilities. However, their behavior in formally structured, rule-based domains remains insufficiently understood. This work investigates the specialization of LLaMA 3.1 for question answering on the Comprehensive Rules of Magic: The Gathering through parameter-efficient fine-tuning using Low-Rank Adaptation (LoRA). The study combines a theoretical analysis of probabilistic language modeling with an empirical evaluation over 100 held-out samples, using BLEU, ROUGE-L, and BERTScore (F1) to capture lexical, structural, and semantic alignment. Although the evaluation set is limited to 100 samples for computational efficiency, the observed trends are consistent across metrics and supported by qualitative analysis. Results show consistent improvements across all metrics, accompanied by reduced variance, indicating successful distributional adaptation. However, a qualitative analysis reveals persistent hallucinations, particularly in the generation of incorrect rule identifiers, highlighting a critical limitation: improvements in standard text similarity metrics do not necessarily correspond to correct rule-grounded reasoning. These findings suggest that LLMs, even after fine-tuning, remain fundamentally probabilistic systems that approximate linguistic patterns rather than reliably encoding symbolic knowledge. This work therefore contributes not only an empirical evaluation of domain adaptation in a structured regulatory setting, but also a broader reflection on the limitations of current evaluation methodologies and the challenges of applying LLMs to formally constrained domains.

1 Introduction

The rapid evolution of Large Language Models has fundamentally transformed computational linguistics and artificial intelligence. Architectures based on the transformer paradigm now achieve human-like fluency in generation, translation, summarization, and question answering. These achievements are rooted in large-scale self-supervised training over heterogeneous corpora, enabling models to internalize statistical regularities across diverse linguistic domains.

However, the deployment of LLMs in highly formalized and rule-governed domains introduces conceptual and methodological challenges. Unlike open-domain dialogue, regulatory corpora impose strict semantic constraints. Precision is not merely stylistic but normative: incorrect phrasing may alter the legal or logical meaning of a statement.

The Comprehensive Rules of Magic: The Gathering provide a particularly rigorous test case. Magic: The Gathering (colloquially known as Magic or MTG) is a collectible, tabletop, and digital collectible card game created by Richard Garfield. Wizards of the Coast released Magic in 1993 as its first trading card game. As

of February 2023, it has amassed approximately fifty million players From 2008 to 2016, over twenty billion Magic cards were printed as the game grew in popularity. For the 2022 fiscal year, Hasbro announced Magic had generated 1 billion in annual revenue.

Players in a game of Magic represent dueling wizards called Planeswalkers. Each card a player draws from their deck represents a magical spell which can be used to their advantage in battle. Instant and Sorcery cards represent magical spells a player may cast for a one-time effect, while Creature, Artifact, Enchantment, Planeswalker, and Battle cards remain on the Battlefield to provide long-term advantage. Players usually must include resource, or Land cards representing the amount of mana that is available to cast their spells. Typically, a player defeats their opponent(s) by reducing their life totals to zero, which is commonly done via combat damage by attacking with creatures. Many other sources of damage exist in the game, in addition to alternative win-conditions which do not check life totals.

Magic can be played by two or more players, either in person with paper cards or on a computer, smartphone or tablet with virtual cards through Internet-based software. It can be played in various rule formats, which fall into two categories: constructed and limited. Limited formats involve players creating a deck spontaneously out of a pool of random cards typically with a minimum deck size of 40 cards. In constructed formats, players create decks from cards they own, usually with a minimum of 60 cards per deck.

New cards are released on a regular basis through expansion sets. Further developments include the Wizards Play Network played at the international level and the worldwide community Players Tour, as well as a substantial resale market for Magic cards. Certain cards can be valuable due to their rarity in production and utility in gameplay, with prices ranging from a few cents to tens of thousands of dollars.

The MTG rulebook functions as a formal system, containing hierarchical definitions, exception hierarchies, layered effect systems, and cross-referenced clauses. This structured nature renders it analogous to legal codes or technical standards.

The present study aims to investigate whether parameter-efficient domain adaptation can enhance the fidelity of a general-purpose LLM when applied to such a structured environment.

This work makes three primary contributions. First, it provides an empirical evaluation of parameter-efficient fine-tuning for question answering in a highly structured regulatory domain. Second, it demonstrates that improvements in standard similarity-based metrics do not necessarily correspond to improvements in rule-grounded correctness. Third, it offers a qualitative analysis of hallucination patterns, highlighting systematic limitations in the model's ability to map between symbolic identifiers and semantic content.

2 Research Questions

This work is guided by the following research questions:

- To what extent can parameter-efficient fine-tuning improve the performance of a general-purpose LLM in a structured, rule-based domain?
- Do improvements in standard NLP evaluation metrics correspond to improvements in rule-grounded correctness?
- What types of errors persist after domain adaptation, and what do they reveal about the limitations of probabilistic language models?

3 Theoretical Foundations of Probabilistic Language Modeling and Domain Adaptation

Large Language Models are trained through autoregressive objectives. Given a token sequence $x = (x_1, x_2, ..., x_n)$ the model optimizes:

$$\mathcal{L} = -\sum_{t=1}^{n} \log P(x_t | x_{<t})$$

This objective ensures statistical coherence but does not encode symbolic reasoning or explicit logical constraints. Instead, reasoning emerges implicitly through distributed representations learned during training. While this paradigm enables remarkable generalization, it also imposes epistemic limitations, as the model approximates linguistic distributions rather than executing formal inference, a distinction that becomes critical in structured regulatory systems where correctness depends on strict adherence to hierarchical definitions and conditional dependencies. Transformers, which form the architectural backbone of modern LLMs, rely on self-attention mechanisms to model contextual dependencies [1]. Given query, key, and value matrices Q, K, and V, attention is computed as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V$$

Multi-head attention enables parallel contextual projections, allowing the model to capture multiple relational structures simultaneously across different representational subspaces. However, despite their architectural sophistication, attention mechanisms remain fundamentally statistical: they capture correlation patterns present in the training distribution but do not guarantee rule-consistent reasoning or logically valid inference. To address distributional mismatches between pretraining corpora and specialized target domains, domain adaptation techniques are employed. Domain adaptation seeks to adjust a pretrained model so that its internal representations more closely align with the statistical properties of a target distribution. Traditional fine-tuning updates all parameters of the model, which can improve in-domain performance but carries the risk of catastrophic forgetting, potentially degrading previously acquired general knowledge. Parameter-efficient methods such as adapters and Low-Rank Adaptation (LoRA) mitigate this risk by constraining updates to a small subset of additional parameters while keeping the majority of pretrained weights frozen. Prior work has demonstrated the effectiveness of domain adaptation in biomedical and legal corpora, domains characterized by specialized terminology and semi-formal structure; however, comparatively little attention has been given to structured rule-based gaming systems, which share characteristics with legal texts but introduce additional logical layering mechanisms and precedence systems. LoRA introduces trainable low-rank updates to frozen weight matrices in order to achieve efficient specialization. Given a pretrained weight matrix $W \in \mathbb{R}^{d \times k}$, LoRA defines:

$$W' = W + \Delta W$$

where:

$$\Delta W = BA$$

with $B \in \mathbb{R}^{d \times r}$ and $A \in \mathbb{R}^{r \times k}$. The rank r is significantly smaller than d or k, thereby constraining adaptation to a low-dimensional subspace. This constraint can be interpreted geometrically as projecting domain-specific adjustments into a limited representational manifold embedded within the original parameter

space. Such constrained adaptation preserves global linguistic competence while enabling targeted distributional alignment, offering a principled compromise between stability and specialization in formally structured domains.

4 Regulatory Structure of Magic: The Gathering

The Comprehensive Rules document exceeds 200 pages and is organized hierarchically. Rules are indexed numerically (e.g. 704.5, 613.1a), reflecting layered dependencies. These mechanisms resemble formal logical systems and require strict adherence to defined precedence.

Key structural features include the following:

- Replacement and prevention effects
- Continuous effects resolved in layers
- State-based actions
- Priority and timing rules
- Triggered abilities

5 Fine Tuning Dataset Construction and Curation

The "mtg-data" dataset is a collection of prompts and responses related to Magic: The Gathering. The dataset contains various types of question and answer pairs, including official rulings as responses and corresponding questions generated by GPT-3.5, "Question and Answer" data scraped from the web, glossary terms alongside their descriptions, and official rules formatted into Q/A pairs. This dataset is designed to facilitate the development and research of AI models focused on understanding game dynamics, card interactions, and providing judge rulings and is available on huggingface (https://huggingface.co/datasets/nelsntk/mtg-data).

The evaluation metrics have been calculated using 100 held-out samples of the test set to measure generalization while keeping notebook execution length in check.

6 Experimental Setup and Evaluation Metrics

To evaluate model performance, three complementary automatic metrics were employed: BLEU, ROUGE, and BERTScore. Each metric captures a distinct dimension of textual similarity and, when interpreted jointly, provides a comprehensive assessment of generation quality across lexical, structural, and semantic levels.

6.1 BLEU (Bilingual Evaluation Understudy)

BLEU is a precision-oriented metric originally developed for machine translation. It quantifies the degree of lexical overlap between a generated text and one or more human reference texts. Formally, BLEU computes modified n-gram precision, where n-grams are contiguous sequences of n tokens, typically for $n = 1, 2, 3, 4$.

The fundamental intuition behind BLEU is to measure how many of the word sequences produced by the model also appear in the reference answer. Unigram precision evaluates individual word overlap, while

higher-order n-grams capture short phrase similarity and local syntactic conformity. To prevent artificial score inflation due to repetition, BLEU applies clipped counts, ensuring that each n-gram in the generated text is only credited up to the maximum number of times it appears in the reference.

Because a model could artificially achieve high precision by producing extremely short responses containing only highly probable words, BLEU incorporates a brevity penalty (BP). This penalty reduces the score if the generated output is significantly shorter than the reference, thereby discouraging under-generation. The BLEU score is defined as:

$$\text{BLEU} = BP \cdot \exp\left(\sum_{n=1}^{N} w_n \log p_n\right)$$

where p_n denotes modified n-gram precision and w_n are weighting coefficients (typically uniform).

BLEU scores range from 0 to 1 (or equivalently 0 to 100 when scaled). Higher scores indicate stronger lexical alignment with the reference. However, BLEU primarily evaluates surface-level similarity and does not directly assess semantic correctness, logical validity, or discourse coherence. A response may achieve a moderate BLEU score due to lexical overlap even if it contains subtle logical inaccuracies.

In the context of regulatory question answering, BLEU is particularly informative for assessing terminological fidelity—that is, whether the model correctly reproduces domain-specific vocabulary and canonical phrasing patterns.

6.2 ROUGE (Recall-Oriented Understudy for Gisting Evaluation)

ROUGE was originally introduced for summarization evaluation and emphasizes recall rather than precision. While BLEU evaluates how much of the generated text matches the reference, ROUGE measures how much of the reference content has been successfully captured by the model's response.

ROUGE computes the proportion of overlapping units from the reference text that appear in the generated output. This makes it particularly sensitive to completeness and content coverage.

Several variants of ROUGE exist:

- **ROUGE-1**: Measures unigram overlap and captures basic content inclusion.
- **ROUGE-2**: Measures bigram overlap and reflects short-range structural similarity.
- **ROUGE-L**: Measures the Longest Common Subsequence (LCS) between the generated and reference texts.

ROUGE-L is especially relevant in explanatory and regulatory tasks, as it rewards in-order word matching even when words are not contiguous. Because LCS captures global sentence structure rather than fixed-length fragments, it provides insight into discourse organization and argumentative sequencing.

Conceptually, if BLEU emphasizes precision ("Did the model write correct phrases?"), ROUGE emphasizes recall ("Did the model include all essential information?"). In structured rule-based environments, higher ROUGE scores indicate that key rule components, conditions, and constraints have not been omitted.

6.3 BERTScore

BERTScore represents a more recent class of evaluation metrics based on contextual embeddings rather than exact token matching. Unlike BLEU and ROUGE, which depend on surface-level n-gram overlap,

BERTScore computes similarity in semantic embedding space using pretrained transformer encoders such as BERT.

For each token in the generated text, BERTScore identifies the most semantically similar token in the reference text by computing cosine similarity between contextual embeddings. Precision, recall, and F1 variants are then derived from these pairwise similarities.

Given contextual embeddings h_i from the generated text and h_j from the reference text, similarity is computed as:

$$\text{sim}(h_i, h_j) = \frac{h_i \cdot h_j}{\|h_i\| \|h_j\|}$$

This embedding-based approach allows BERTScore to capture synonymy, paraphrasing, and semantic equivalence even when exact lexical overlap is absent. For example, if the model uses a synonymous expression instead of the exact wording found in the reference, BLEU and ROUGE may penalize the output heavily, whereas BERTScore will still assign a high score if semantic meaning is preserved.

In rule-based question answering, BERTScore is particularly valuable because it evaluates conceptual alignment rather than rigid token identity. It therefore provides a closer approximation to human semantic judgment. However, it is important to note that BERTScore, like BLEU and ROUGE, measures similarity rather than formal logical correctness.

6.4 Complementarity of the Metrics

The three metrics evaluate complementary aspects of generation quality. BLEU captures lexical precision and adherence to domain-specific terminology. ROUGE captures coverage and structural completeness, ensuring that essential information is not omitted. BERTScore captures semantic alignment and paraphrastic equivalence at the embedding level.

Taken together, these metrics provide a multi-dimensional evaluation framework. BLEU verifies terminological fidelity, ROUGE verifies informational completeness, and BERTScore verifies semantic consistency. Nevertheless, none of these metrics directly validates formal logical reasoning or rule-consistent inference. They measure similarity to a reference answer, not strict regulatory correctness. This distinction is particularly relevant in structured domains, where surface similarity does not necessarily guarantee formal validity.

6.5 Limitations of Automatic Evaluation

While BLEU, ROUGE, and BERTScore provide useful proxies for lexical, structural, and semantic similarity, they do not capture formal correctness in rule-based systems. In particular, they are insensitive to errors involving incorrect rule references or subtle violations of logical dependencies. As a result, a response may achieve high scores across all metrics while still being factually incorrect. This limitation is especially critical in regulatory domains, where correctness depends on exact symbolic alignment rather than approximate semantic similarity.

7 Results

Improvements are consistent and moderate in magnitude, with variance reduction indicating increased stability across samples. The largest improvement occurs in ROUGE-L, suggesting enhanced structural conformity.

Table 1: Baseline vs Fine-Tuned Performance (n = 100)

Metric	Baseline	Fine-Tuned	Δ
BLEU	0.312 ± 0.048	0.358 ± 0.041	+0.046
ROUGE-L	0.421 ± 0.052	0.487 ± 0.044	+0.066
BERTScore (F1)	0.842 ± 0.027	0.879 ± 0.021	+0.037

BERTScore improvements indicate better semantic fidelity. These gains suggest improved distributional alignment rather than genuine symbolic reasoning. More generally, these metrics capture similarity rather than formal correctness. A response may achieve high semantic similarity while subtly misapplying a rule. Thus, metric improvement should be interpreted as probabilistic alignment rather than formal verification.

8 Analysis of Hallucination and Referential Errors

While the fine-tuned model demonstrated measurable improvements across automatic evaluation metrics, a qualitative inspection of its outputs revealed a persistent and systematic limitation: the hallucination of incorrect rule identifiers when explicitly referencing sections of the Comprehensive Rules. This phenomenon was observed even in otherwise fluent and semantically coherent answers, suggesting that metric improvements did not translate into fully reliable domain-grounded reasoning.

```
You are a Magic: The Gathering judge and rules expert. Answer clearly and precisely.

User:
How does the commander tax work?

Assistant:
The commander tax is a tax that may be paid when a player casts a commander card.
See rule 702.18, "Commander Tax."
```

In the example above, the model-generated reference to “rule 702.18, ”Commander Tax”” is incorrect. The appropriate rule governing the commander tax is Rule 903.8 of the Magic: The Gathering Comprehensive Rules, which states: “A commander cast from the command zone costs an additional 2 for each previous time the player casting it has cast it from the command zone that game.” The fabricated citation illustrates a classical hallucination pattern: the model produces a syntactically plausible and structurally well-formed rule reference, yet one that does not correspond to the actual regulatory structure of the domain.

Notably, such errors were not isolated occurrences. Across multiple evaluation samples, the model exhibited a tendency to generate rule identifiers that were numerically coherent and stylistically consistent with the official ruleset, but semantically misaligned with the queried concept. This behavior suggests that the model internalized statistical regularities regarding rule formatting (e.g., three-digit hierarchical identifiers) without developing a stable mapping between specific rule numbers and their corresponding semantic content.

A plausible explanation for this phenomenon lies in a distributional imbalance in the training data. If the fine-tuning corpus disproportionately emphasized certain rule families or frequently referenced specific sections, the model may have overfit to these patterns, subsequently inferring a limited subset of familiar

rule identifiers in contexts where different citations were required. Rather than retrieving or reconstructing the correct rule index, the model appears to perform a form of probabilistic pattern completion, generating the most statistically plausible rule reference given the learned distribution.

This observation underscores an important methodological consideration: improvements in surface-level similarity metrics such as BLEU, ROUGE, or BERTScore do not necessarily imply improvements in factual grounding or referential precision. In highly structured regulatory domains, correctness depends not only on semantic coherence but also on exact symbolic alignment. Consequently, future work should incorporate targeted evaluation protocols specifically designed to measure citation accuracy and rule-level factual consistency, complementing traditional text similarity metrics.

More broadly, this behavior can be interpreted as a failure of binding between symbolic identifiers and semantic representations. While the model successfully learns the syntactic structure of rule references, it does not reliably associate specific identifiers with their corresponding meanings. This suggests that the underlying representation remains distributional rather than symbolic, limiting its applicability in domains where precise referential accuracy is required.

9 Future Directions

The results of this study highlight several promising directions for future research, particularly in addressing the limitations observed in factual grounding and rule-specific hallucinations. A first natural extension concerns the expansion and diversification of the training corpus. While the current dataset is primarily derived from structured rule explanations, incorporating additional sources of real-world question answering data may significantly improve the model's robustness and generalization capabilities. In this regard, community-driven platforms represent a valuable yet underutilized resource. Online forums such as specialized Magic: The Gathering subreddits (e.g., `/r/mtgrules` and `/r/askajudge`) contain large volumes of organically generated question-answer pairs that reflect practical gameplay scenarios, edge cases, and informal explanatory styles. Scraping and curating such data could enrich the training distribution with more diverse linguistic patterns and real-world reasoning contexts, thereby reducing overfitting to overly formal or templated rule descriptions.

However, increasing dataset size alone is unlikely to resolve the issues identified in this work. A second, and arguably more critical, direction involves systematic dataset cleaning and restructuring. The observed hallucinations in rule identifiers suggest that the model has learned superficial correlations between question types and frequently occurring rule numbers, rather than a grounded mapping between rules and their semantic content. This indicates that the presence of explicit rule references in the training data may inadvertently encourage the model to generate plausible but incorrect citations. Given that real-world judge interactions in Magic: The Gathering rarely require citing exact rule numbers, but instead focus on clearly resolving gameplay situations, it may be beneficial to remove or de-emphasize rule identifiers during training. By shifting the objective toward accurate explanation rather than explicit citation, the model can be encouraged to prioritize semantic correctness over symbolic recall.

In addition, future work could explore more principled data curation strategies, such as filtering ambiguous or low-quality examples, balancing the distribution of rule categories, and explicitly annotating edge cases involving complex interactions (e.g., layered effects or replacement rules). Another promising direction involves the integration of retrieval-based mechanisms, allowing the model to access the official rule corpus at inference time rather than relying solely on parametric memory. Such approaches could reduce hallucinations

by grounding responses in verifiable sources.

More broadly, these directions point toward a hybrid paradigm in which large language models are not treated as self-contained knowledge systems, but as components within a larger pipeline combining curated data, retrieval, and domain-specific constraints. Advancing in this direction would not only improve performance in the Magic: The Gathering domain, but also contribute to the development of more reliable AI systems for structured, rule-based environments.

10 Conclusion

This work set out to investigate whether parameter-efficient fine-tuning could improve the performance of a general-purpose Large Language Model in a highly structured, rule-based domain, namely the Comprehensive Rules of Magic: The Gathering. The results provide a nuanced answer. From a quantitative perspective, fine-tuning via LoRA yielded consistent improvements across all evaluated metrics, including BLEU, ROUGE-L, and BERTScore. These gains indicate enhanced lexical alignment, improved structural completeness, and stronger semantic correspondence with reference answers. Moreover, the observed reduction in variance suggests that the fine-tuned model exhibits more stable behavior across different inputs, pointing to a successful adaptation at the level of distributional alignment.

However, a deeper qualitative analysis reveals that these improvements remain fundamentally limited in scope. In particular, the persistence of hallucinated rule identifiers highlights a critical gap between surface-level similarity and true domain understanding. While the model demonstrates the ability to reproduce stylistically and syntactically plausible answers, it does not reliably ground its outputs in the underlying symbolic structure of the rule system. This discrepancy reflects a broader limitation of probabilistic language models: they approximate statistical patterns in data rather than performing formal reasoning or symbolic inference.

The findings of this study therefore reinforce an important conceptual distinction. Performance gains measured through standard NLP metrics should be interpreted as improvements in probabilistic alignment with reference distributions, not as evidence of genuine rule-consistent reasoning. In structured regulatory domains, correctness is inherently tied to precise symbolic mappings and hierarchical dependencies, which are only partially captured by current evaluation frameworks.

At the same time, the results demonstrate that parameter-efficient methods such as LoRA provide a viable and computationally efficient pathway for domain specialization. The ability to improve performance without modifying the full parameter space suggests that much of the required adaptation lies in relatively low-dimensional adjustments to the model's representation space. This insight has practical implications for deploying LLMs in specialized domains where computational resources and data availability may be constrained.

More broadly, this work contributes to the growing body of literature examining the limits of Large Language Models in formalized knowledge systems. The Magic: The Gathering rules corpus serves as a compelling intermediate domain between natural language and formal logic, exposing both the strengths and weaknesses of current approaches. The observed behavior suggests that future progress will likely depend on hybrid architectures that combine the generative flexibility of LLMs with mechanisms for retrieval, verification, or symbolic reasoning.

In conclusion, while fine-tuning improves the linguistic and semantic quality of model outputs, it does not, in its current form, resolve the fundamental challenge of grounding language generation in formally

structured knowledge. Addressing this gap remains an open problem and a critical direction for future research in the development of reliable AI systems for rule-based environments.

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